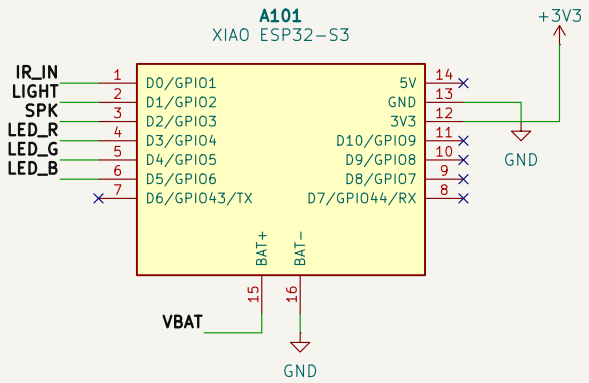
CamoBots — HIDER BODY

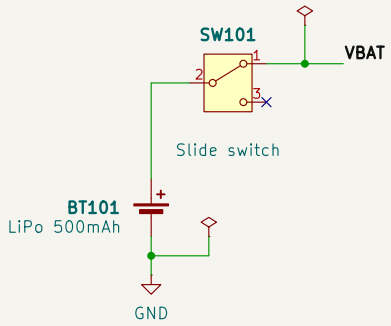
IR receiver catches the blaster's shot > light sensor catches cheating cover > speaker whistles > RGB LED shows the state

HOST (XIAO ESP32-S3)



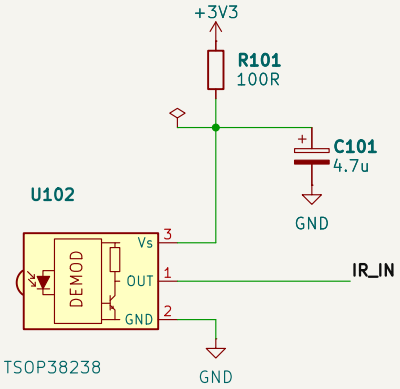
D0–D5 are ADC1 pins, safe to read with Wi-Fi on. D2 is a boot strapping pin: the 10k pull-down keeps it low. If boot misbehaves, move the speaker to D8 (GPIO7).

BATTERY + SWITCH



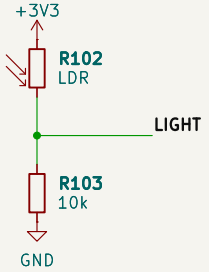
The switch sits in the battery + lead. VBAT is the battery after the switch and goes to the BAT+ pad under the XIAO. Charge through the XIAO USB-C port. The 5V pin is dead on battery. Fit the battery and switch last.

IR RECEIVER



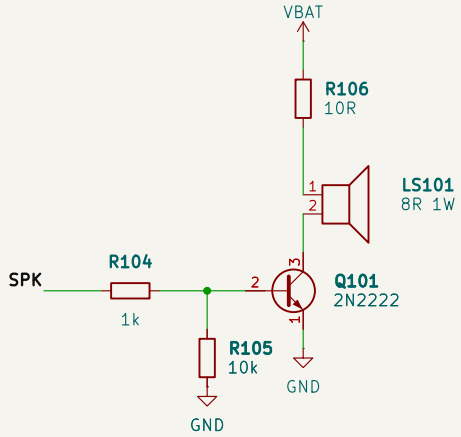
OUT idles HIGH and pulses LOW on a 38 kHz burst. 100R + 4.7u filter its supply. Dome facing you, legs down: OUT GND VS. Never paint over the dome.

LIGHT SENSOR



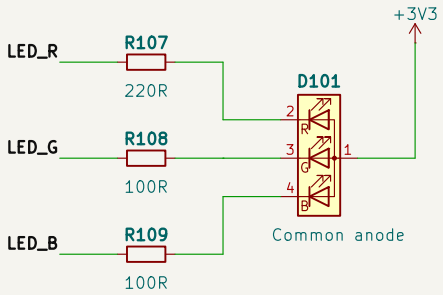
More light = lower LDR resistance = higher reading on D1. Room light about 900, covered about 100. Below 40% of the start level counts as covered.

SPEAKER DRIVER



D2 HIGH switches the transistor on and the speaker sounds. 1k limits the base current; 10k holds it off while the XIAO boots. 10R keeps the speaker current safe. Speaker + goes to VBAT. 2N2222 / S8050: flat face to you, legs down = E B C (check yours).

RGB STATUS LED



The longest leg is the common anode: it goes to 3V3. A pin driven LOW turns its colour on. Red gets a bigger resistor because it is brighter. Teal = waiting, blue = hiding, red = caught, rainbow = survived the round.

PARTS (ONE BODY)

A101	Seeed XIAO ESP32-S3 + 2.4 GHz antenna
U102	TSOP38238 38 kHz IR receiver
R102	LDR 5 mm (light sensor)
Q101	2N2222 or S8050 NPN transistor
LS101	Speaker 8 ohm 1 W (TR-WS-2014B)
D101	5 mm RGB LED, common anode
BT101	LiPo 3.7 V 400-500 mAh, protected
SW101	Slide switch
C101	4.7 uF electrolytic
R101 R108 R109	100R
R103 R105	10k
R104	1k
R106	10R
R107	220R

Drawn by Abhinav Krishna — CamoBots
Sheet 1 of 2 | designators 1xx
TSOP38238 + LDR + 2N2222 speaker driver + common-anode RGB LED
Single-cell LiPo, everything runs from the XIAO 3V3 pin except the speaker
CamoBots | Designed by Abhinav Krishna

Sheet: /
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Title: Hider Body

Size: A4	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/1